

(12) **United States Patent**
Matsui et al.

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(54) **PRESSURE SENSOR DEVICE, METHOD FOR MANUFACTURING THE SAME, AND WORK MANAGEMENT SYSTEM USING THE SAME**

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(30) **Foreign Application Priority Data**

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G06F 3/01 (2006.01)
G01L 1/20 (2006.01)
G01L 1/22 (2006.01)

(52) **U.S. Cl.**
CPC **G06F 3/014** (2013.01); **G01L 1/205** (2013.01); **G01L 1/225** (2013.01)

(58) **Field of Classification Search**
CPC G06F 3/014; G01L 1/205; G01L 1/225; G01L 1/18; G01L 5/0028
See application file for complete search history.

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(57) **ABSTRACT**

A pressure sensor device, a method for manufacturing the pressure sensor device, and a work management system mitigating a degree of a false detection is presented. The pressure sensor device detecting pressure includes a flexible substrate base material having flexibility; a comb-teeth shape electrode having an exposed metal surface formed in a predetermined area on the flexible substrate base material; and a pressure-sensitive material that is provided on the comb-teeth shape electrode, varies in a resistance value depending on an amount of a load, and has a curvature in a static state.

11 Claims, 19 Drawing Sheets

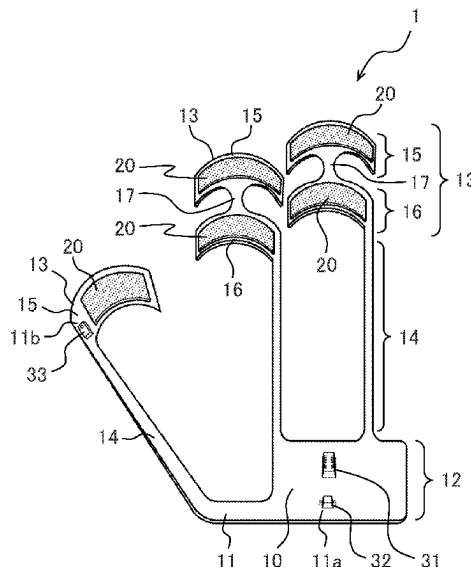


FIG. 1

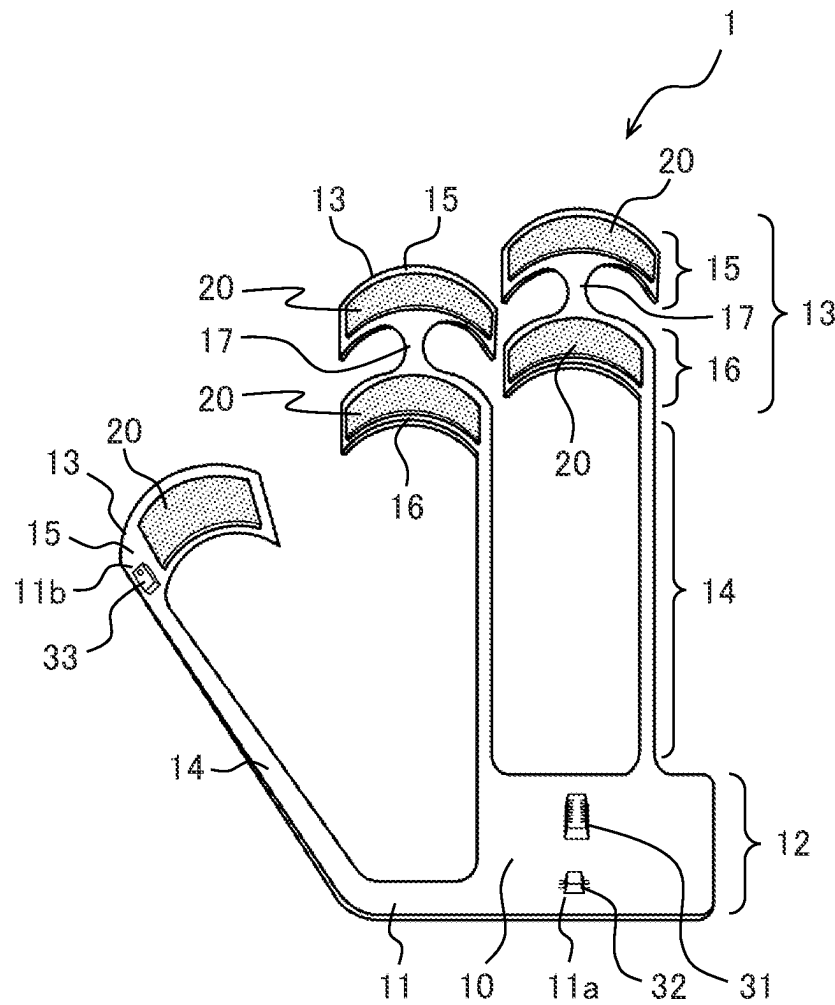


FIG. 2

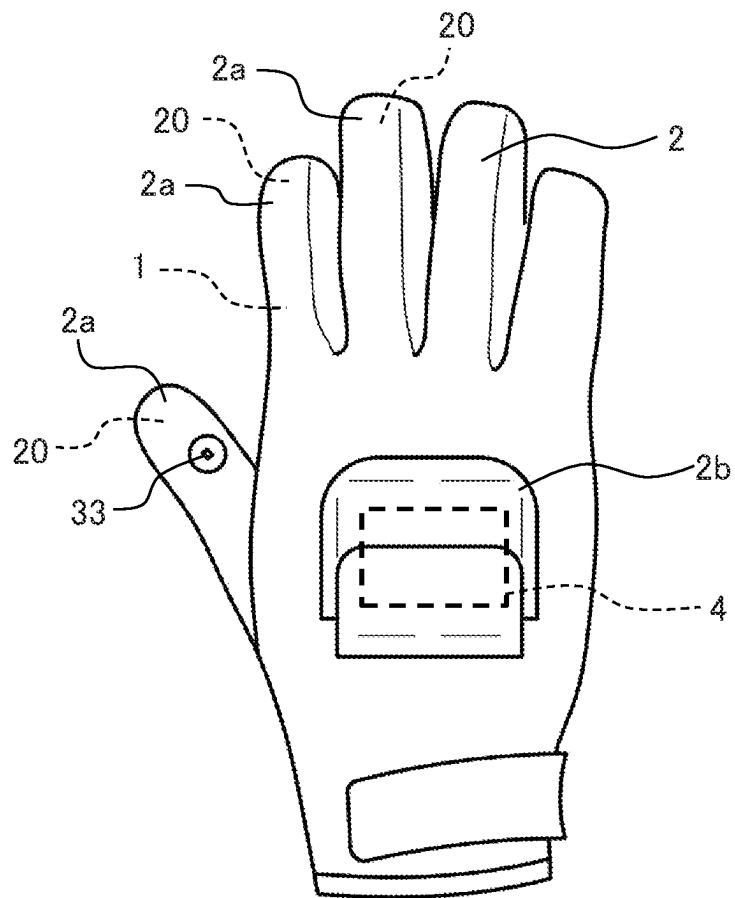


FIG. 3

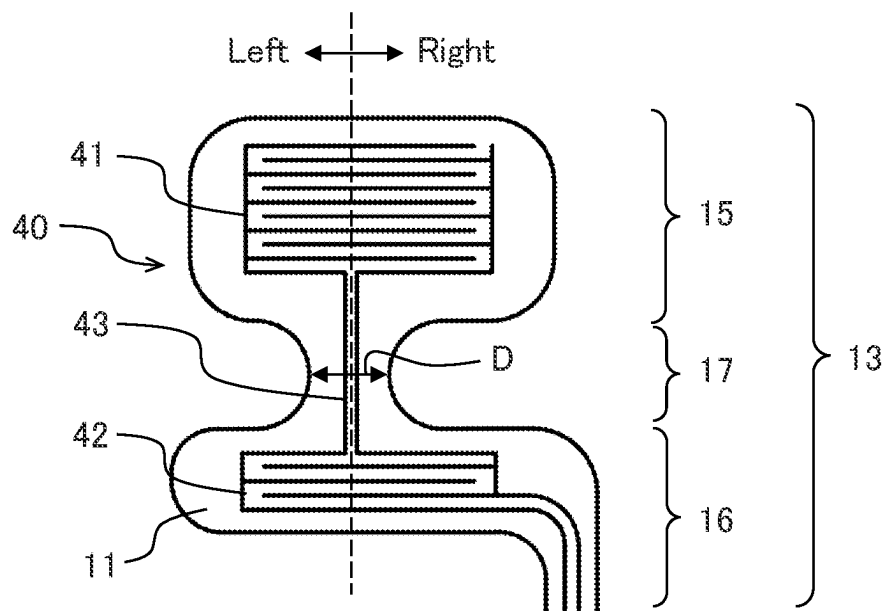


FIG. 4

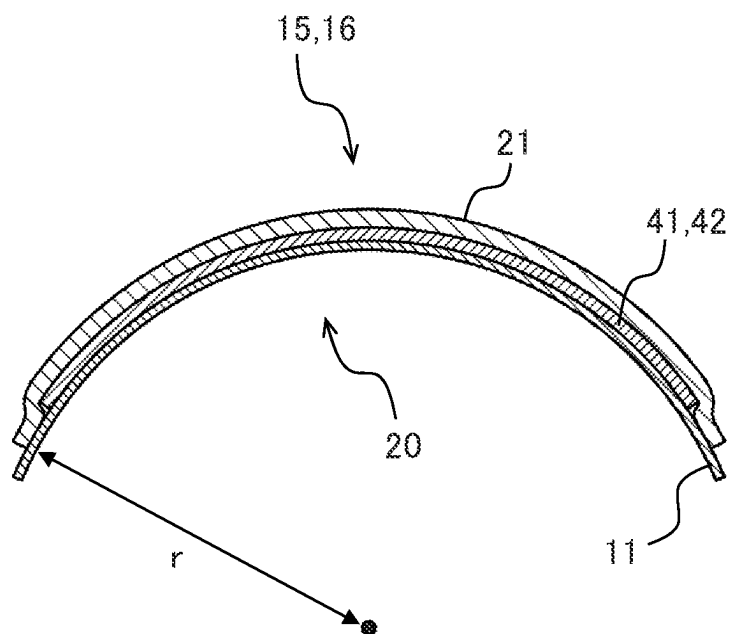


FIG. 5

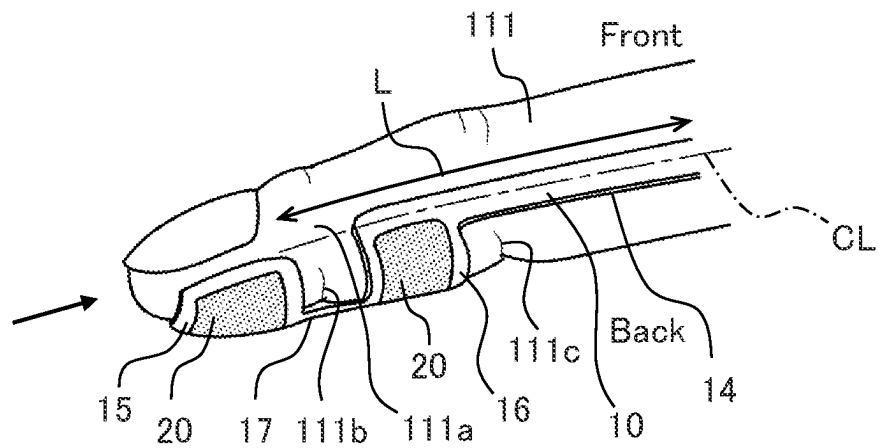


FIG. 6

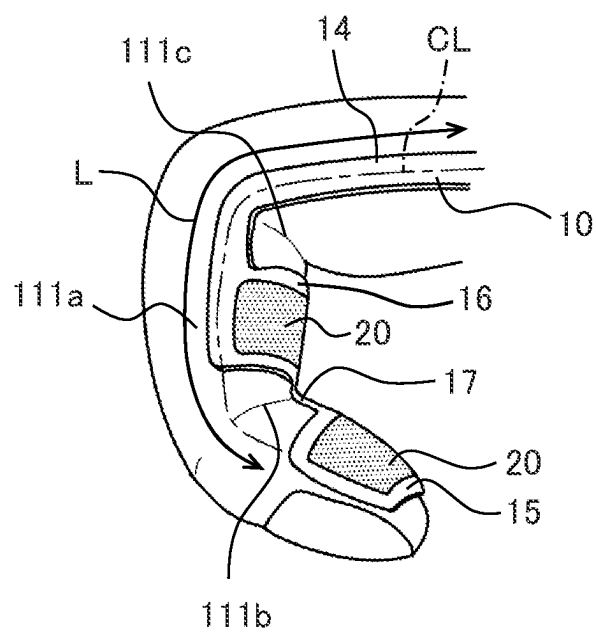


FIG. 7

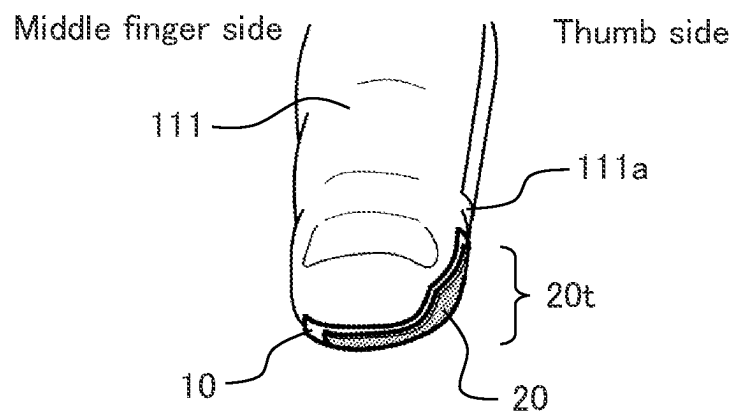


FIG. 8

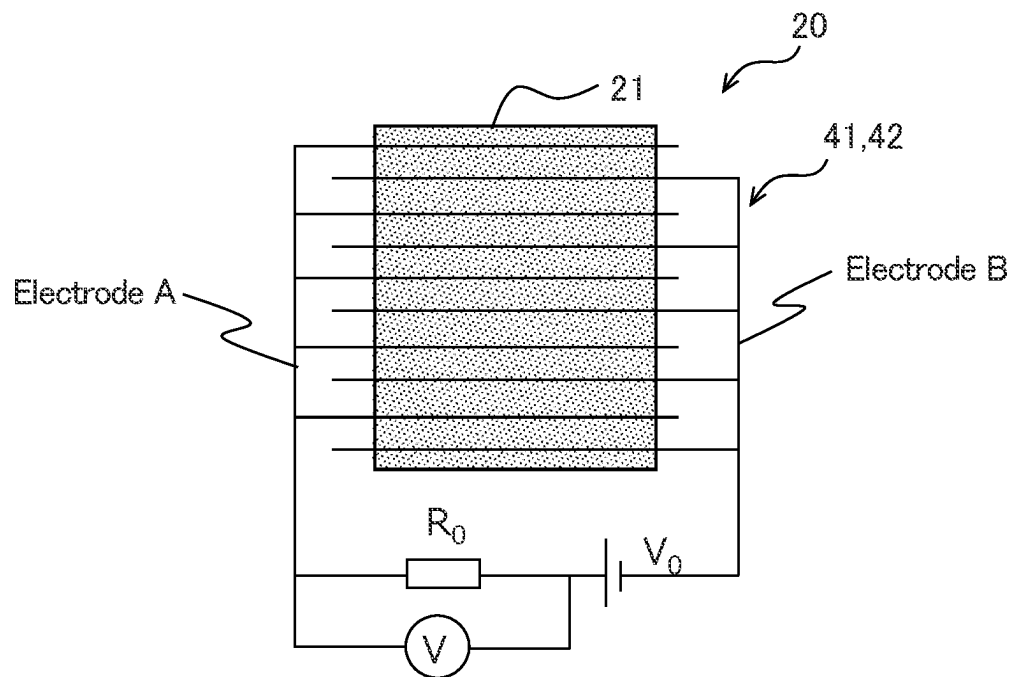


FIG. 9

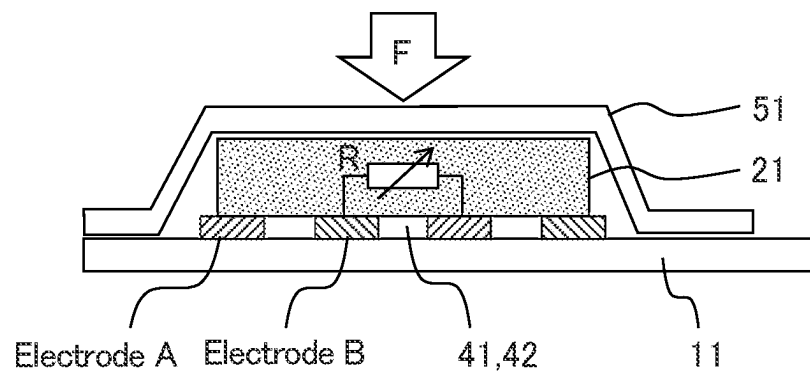
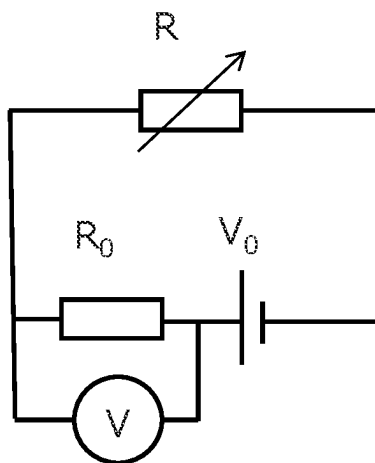


FIG. 10



$$V = \frac{R_0}{R + R_0} V_0$$

FIG. 11

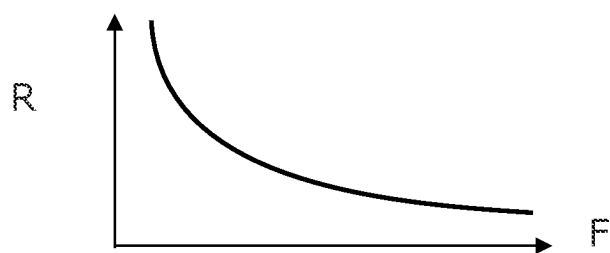


FIG. 12



FIG. 13A

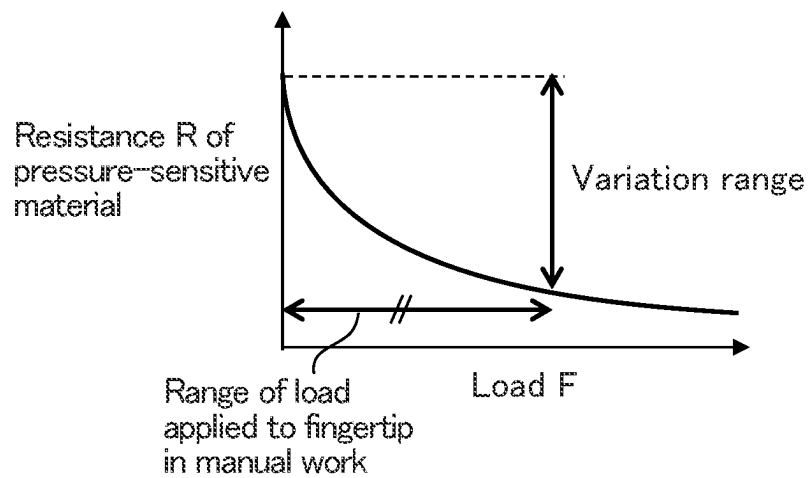


FIG. 13B

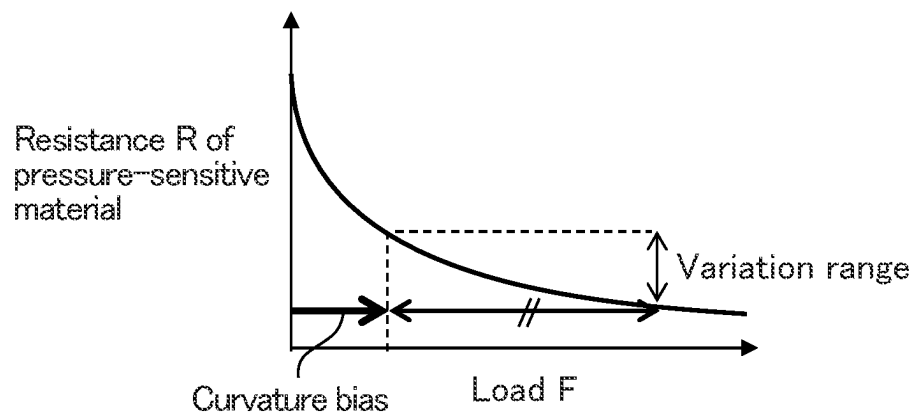


FIG. 14A

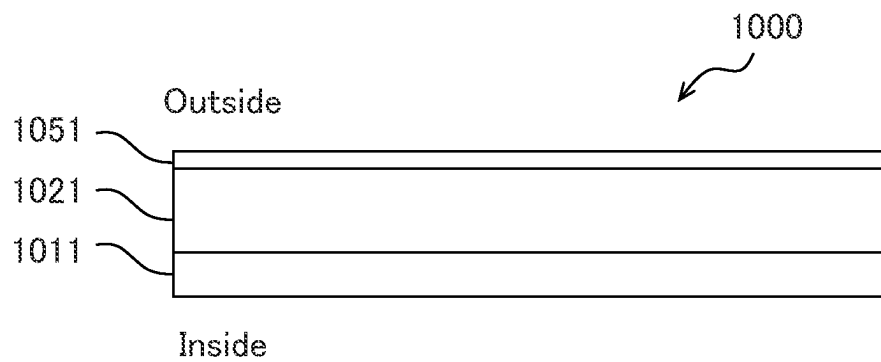


FIG. 14B

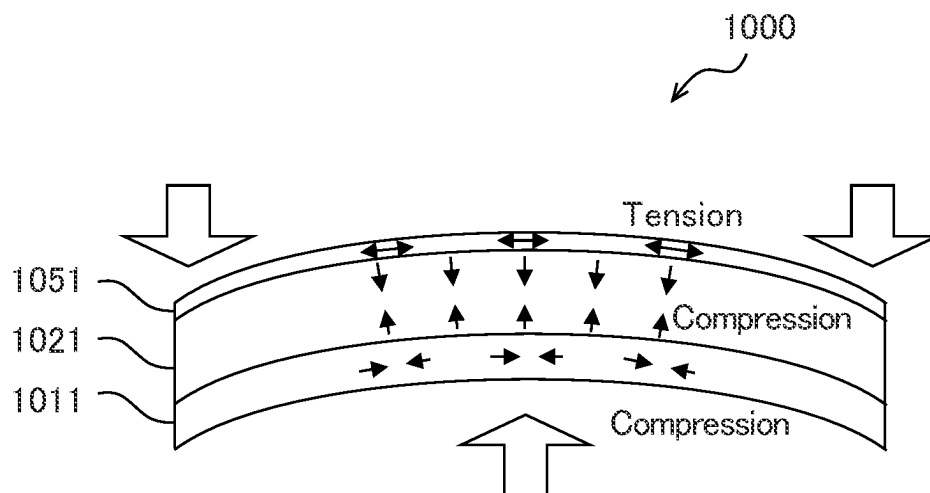


FIG. 15A

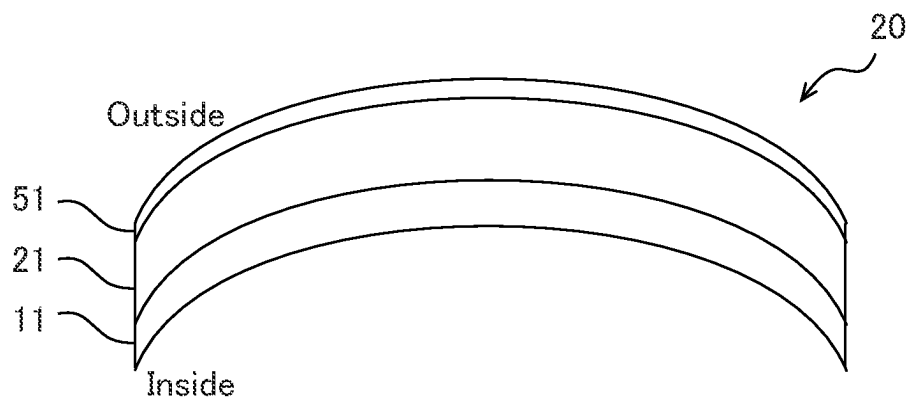


FIG. 15B

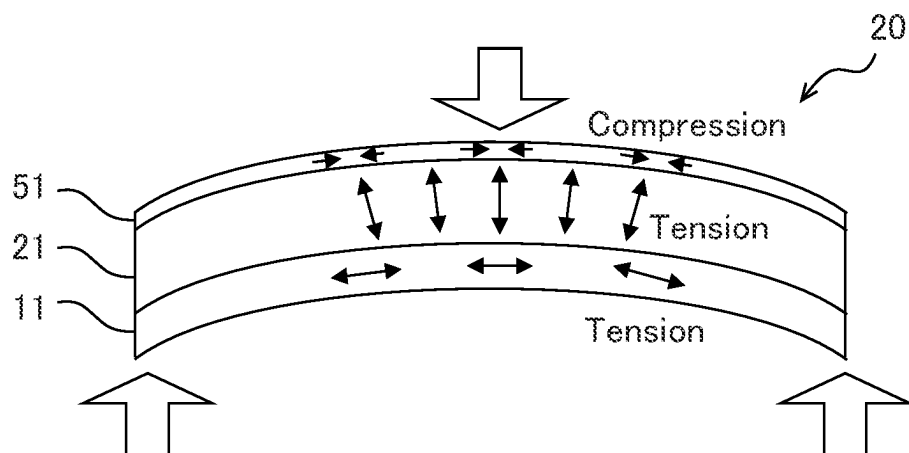


FIG. 16

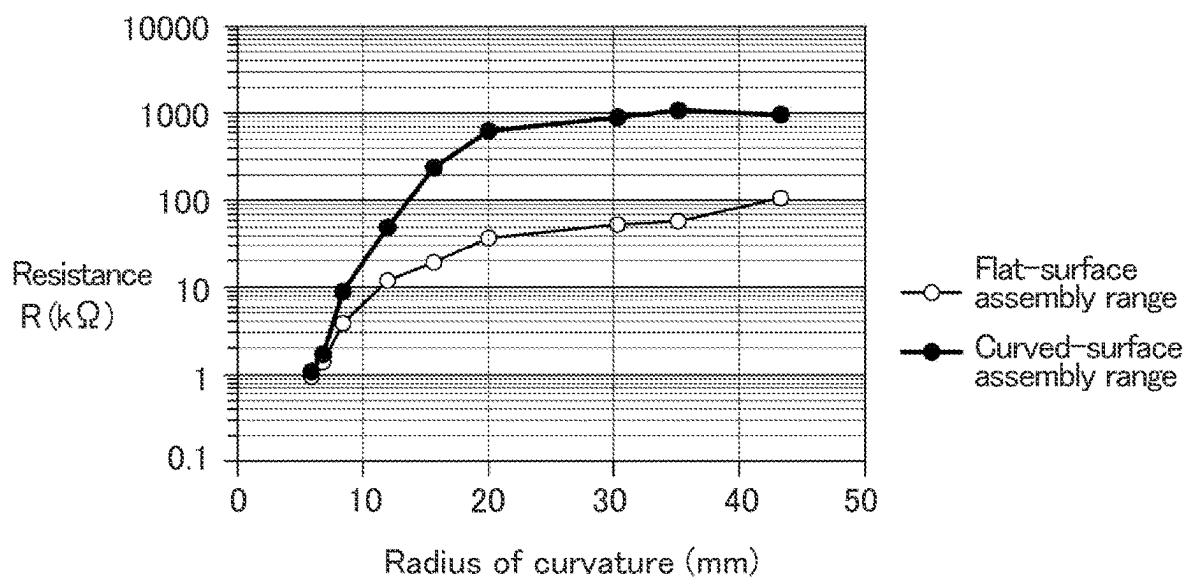


FIG. 17

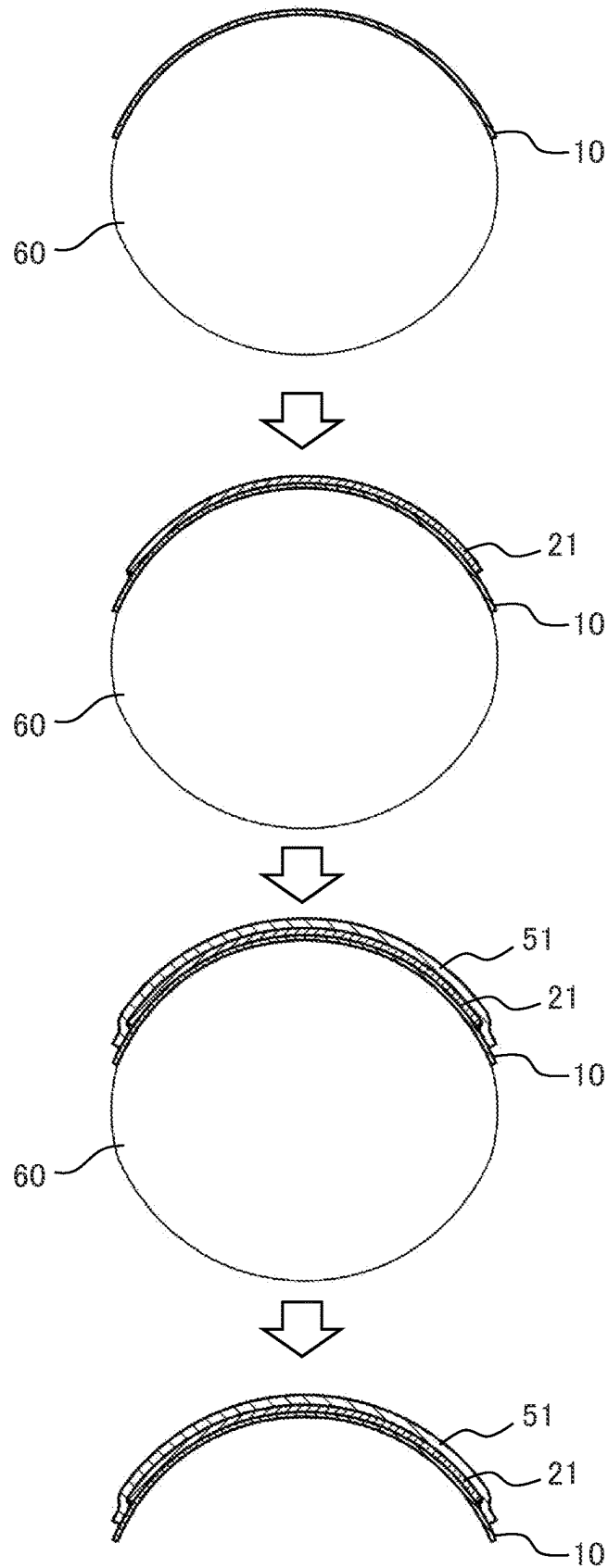


FIG. 18

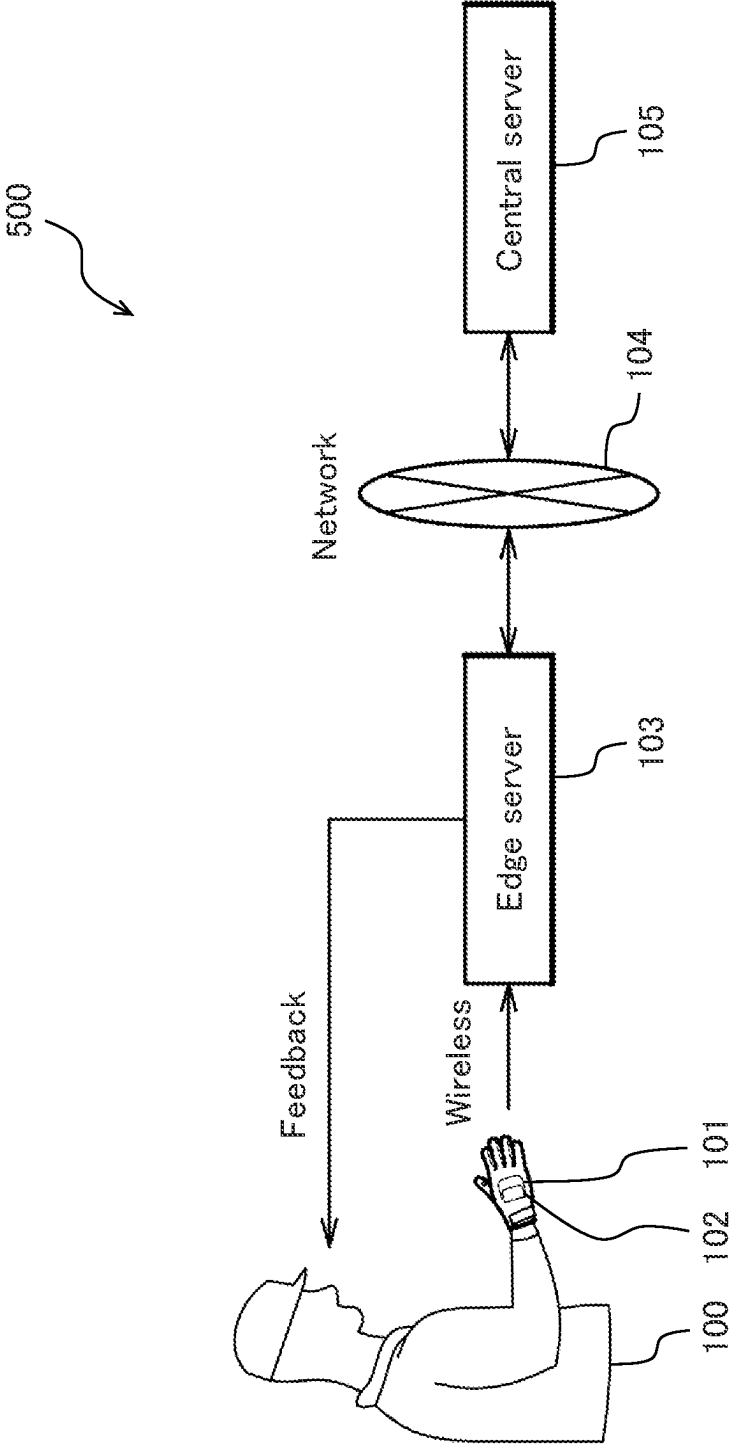


FIG. 19

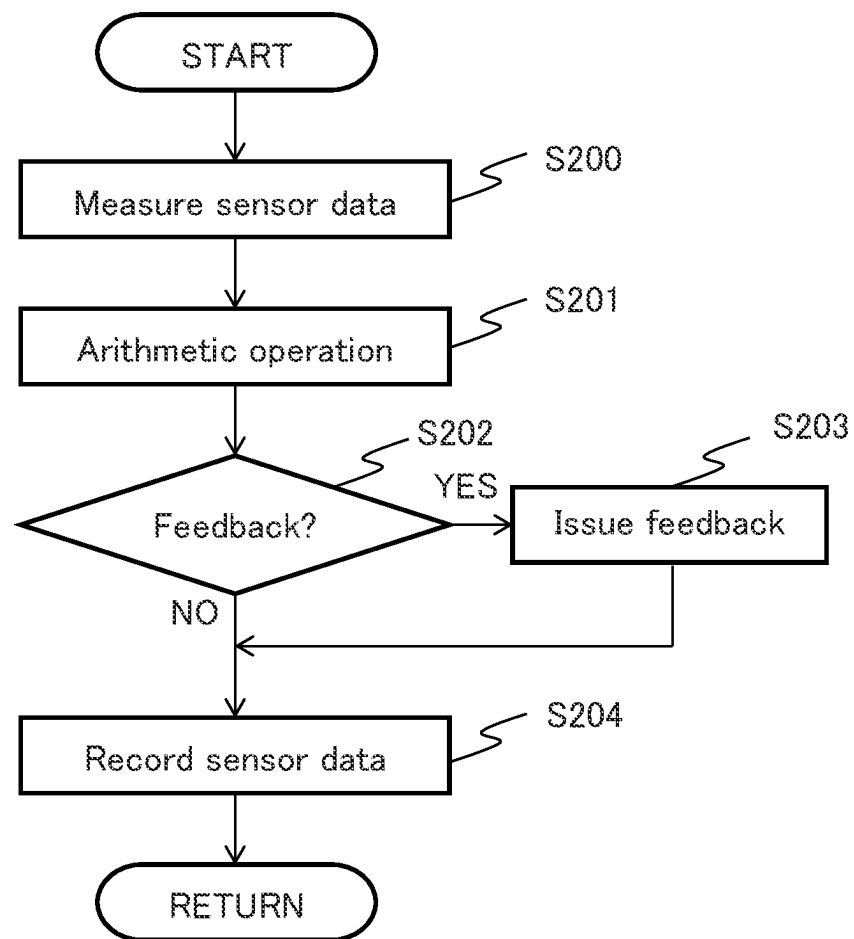


FIG. 20

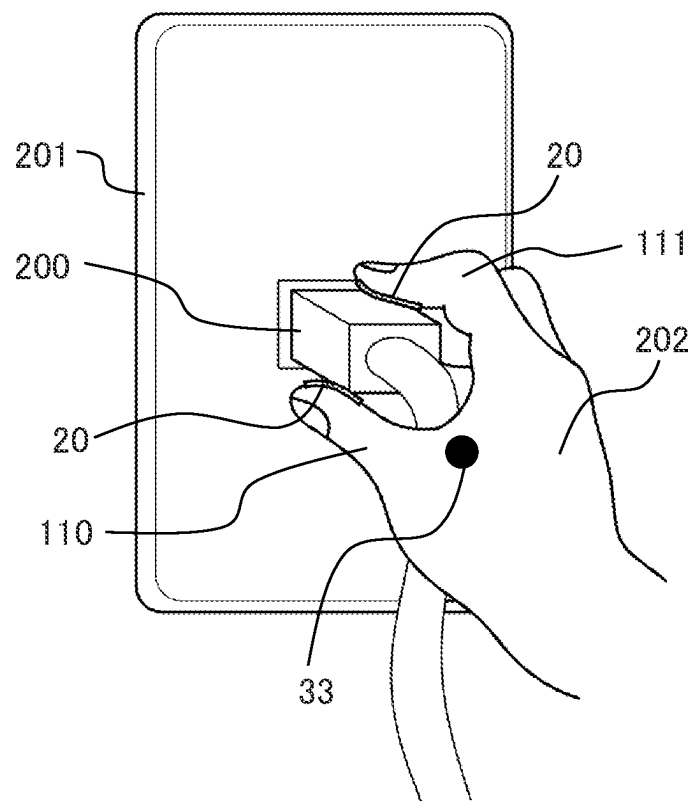


FIG. 21A

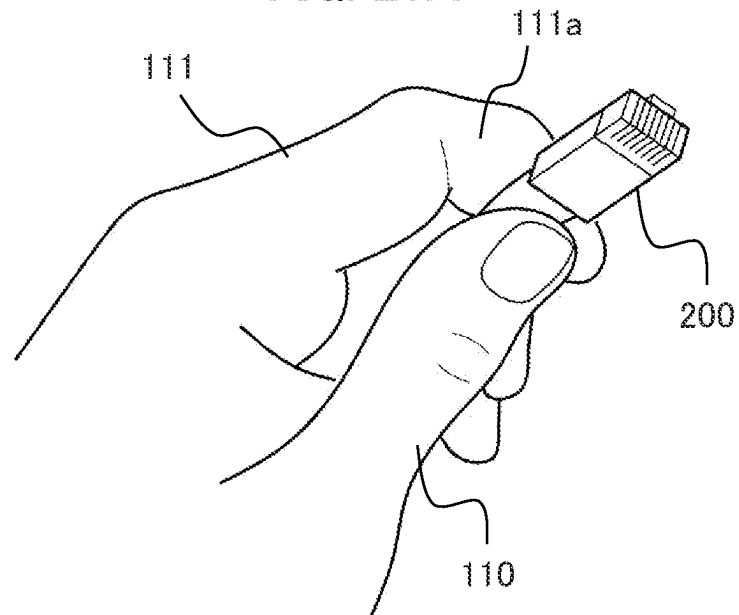


FIG. 21B

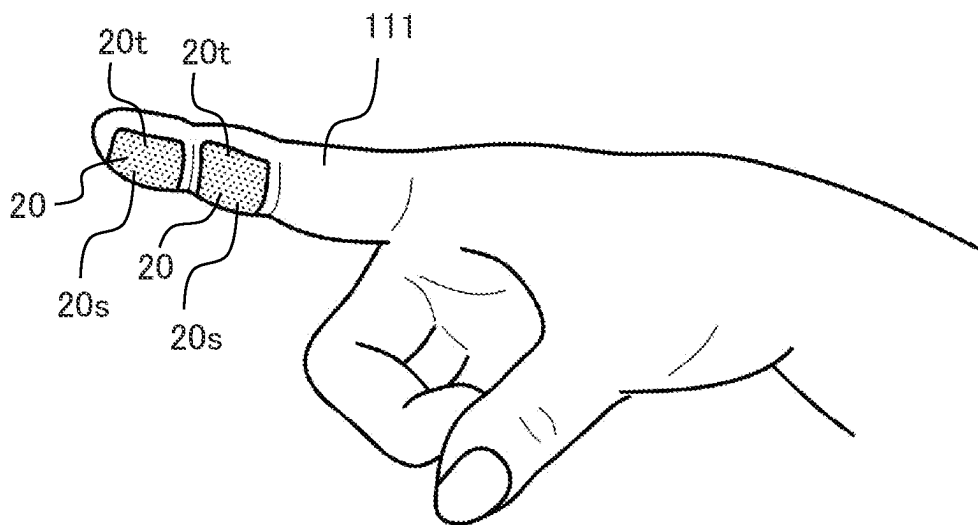


FIG. 22

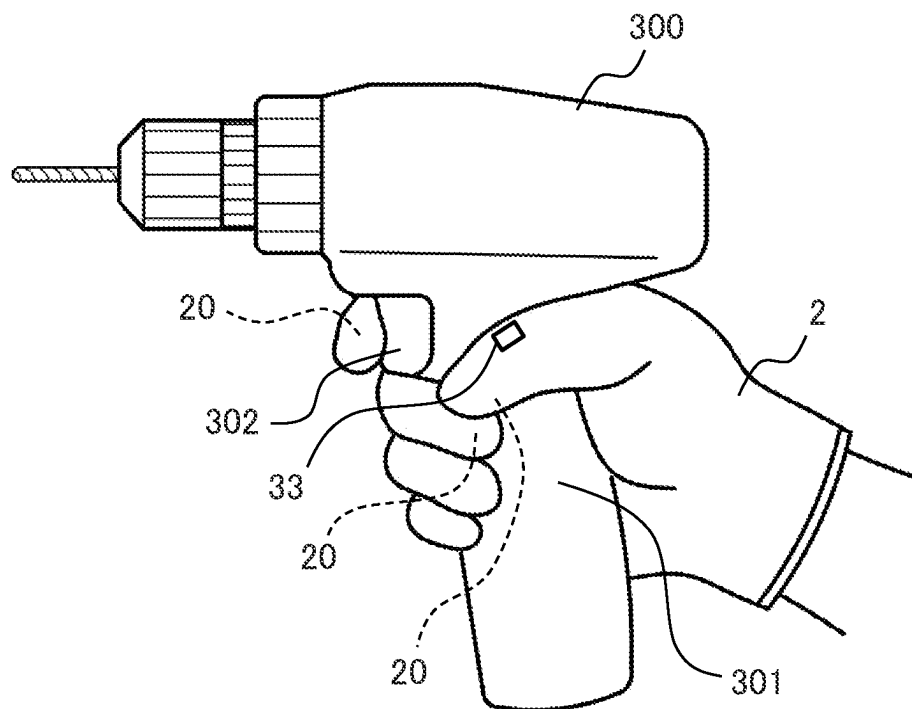


FIG. 23

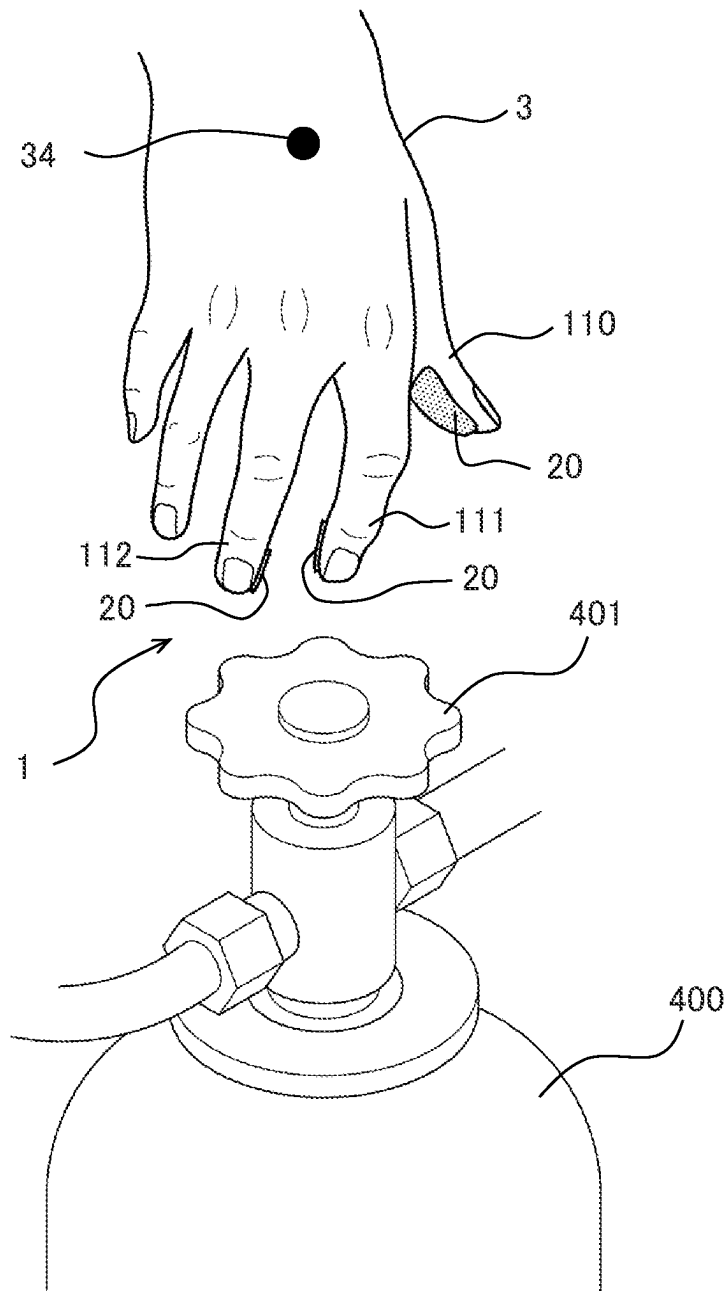
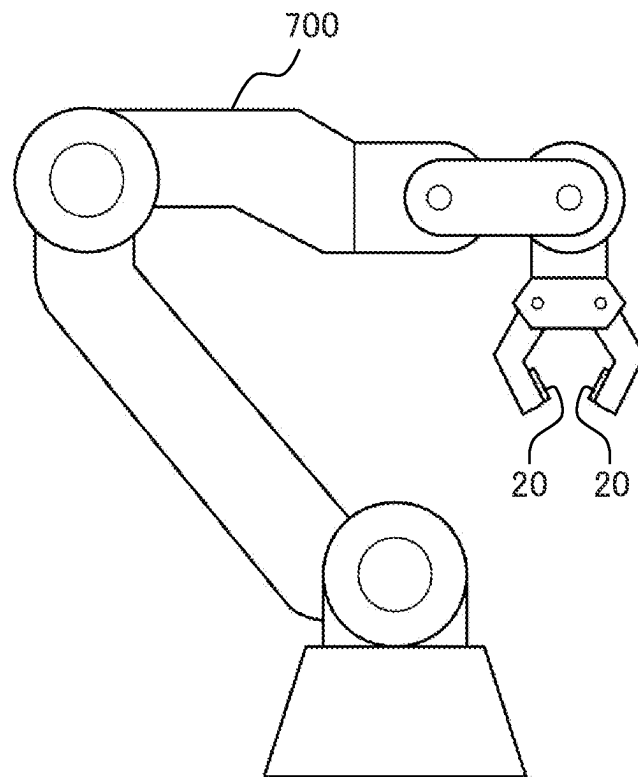


FIG. 24



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PRESSURE SENSOR DEVICE, METHOD FOR MANUFACTURING THE SAME, AND WORK MANAGEMENT SYSTEM USING THE SAME

CROSS-REFERENCE TO RELATED APPLICATIONS

The present invention relates to and asserts priority from Japanese patent application No. 2021-072739 filed on Apr. 22, 2021, and incorporates the entirety of the contents and subject matter of all the above application herein by reference.

TECHNICAL FIELD

The present invention relates to a pressure sensor device, a method for manufacturing a pressure sensor device, and a management system for work using the pressure sensor device.

BACKGROUND

PTL 1 (Patent Literature 1) describes a sensor that reads a load value using a curvature bias load generated by a piezoelectric sensor being curved when the load is applied. PTL 2 (Patent Literature 2) describes a bridge circuit consisting mainly of four resistors R1 to R4 as pressure-sensitive elements.

CITATION LIST

Patent Literature

[PTL 1] Japanese Patent Publication No. 2020-165874
[PTL 2] Japanese Patent Publication No. 2007-10383

SUMMARY OF THE INVENTION

Technical Problem

Recently, it is expected that a sensor is utilized to digitize a manual work to record a work history, to prevent a work error, and to extract a proficient work skill in assembly process and equipment inspection in a factory. In particular, it is important to sense a pressure caused by a force applied on fingertips in the manual work, and therefore a glove with a built-in pressure sensor is required.

However, when the piezoelectric sensor described in PTL 1 is attached to a glove, bending stress is applied to a pressure-sensitive material, which causes an incorrect response of the sensor.

In the sensor described in PTL 2, a flexible substrate on which the pressure-sensitive element is mounted is affixed to a curved surface, but affixing the flexible substrate to the curved surface generates a curvature bias load, which causes detection sensitivity to be impaired.

Accordingly, it is an object of the present invention to provide a pressure sensor device that is capable of mitigating a degree of a false detection, a method of manufacturing the pressure sensor device, and a management system for a work using the pressure sensor device.

Solution to Problem

The present invention provides a pressure sensor device for detecting pressure, the device comprising a flexible

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substrate base material having flexibility; a comb-teeth shape electrode having an exposed metal surface and formed in a predetermined area on the flexible substrate base material; and a pressure-sensitive material that is provided on the comb-teeth shape electrode, varies in a resistance value thereof depending on an amount of a load, and has a curvature in a static state.

BRIEF DESCRIPTION OF DRAWINGS

FIG. 1 is a perspective view of a pressure sensor device according to an embodiment.

FIG. 2 is an external perspective view of the pressure sensor device attached to a glove.

FIG. 3 is a plan view showing a part of a flexible substrate used in the pressure sensor device of the embodiment.

FIG. 4 is a cross-sectional view of the pressure sensor device.

FIG. 5 is a perspective view of a state in which the pressure sensor device is attached to a finger and the finger is extended.

FIG. 6 is a perspective view showing a state of the fingers bent from the state shown in FIG. 5.

FIG. 7 is an arrow view in a S direction of FIG. 5.

FIG. 8 is a circuit diagram of a reading circuit reading a resistance of a pressure-sensitive material using a comb-teeth shape electrode.

FIG. 9 is a cross-sectional schematic diagram of the pressure sensor.

FIG. 10 is a circuit diagram of the pressure sensor.

FIG. 11 is a R-F characteristics of the pressure sensor.

FIG. 12 is a V-F characteristics of the pressure sensor.

FIG. 13A is a F-R characteristics of the pressure-sensitive material without a curvature bias.

FIG. 13B is a F-R characteristics of the pressure-sensitive material with a curvature bias.

FIG. 14A is a cross-sectional view of the pressure sensor fabricated by a flat-surface assembly.

FIG. 14B is a schematic diagram showing an internal stress when the pressure sensor of FIG. 14A is curved.

FIG. 15A is a cross-sectional view of the pressure sensor fabricated by a curved-surface assembly.

FIG. 15B is a schematic diagram showing an internal stress when a curvature of the pressure sensor of FIG. 15A is eliminated.

FIG. 16 is curvature characteristics of the pressure sensors fabricated by a flat-surface assembly and a curved-surface assembly.

FIG. 17 is a process diagram showing a manufacturing method of the pressure sensor device.

FIG. 18 is a configuration diagram of a work management system managing a work performed by a worker.

FIG. 19 is a flowchart showing an operation of the work management system.

FIG. 20 is a perspective view of an example of a connector insertion work.

FIG. 21A is a perspective view showing a shape of a hand holding a connector.

FIG. 21B is a perspective view showing a position of the finger on which the sensor is mounted.

FIG. 22 is a perspective view of an example of a work using a power tool.

FIG. 23 is a perspective view showing an example of a work for opening/closing a valve.

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FIG. 24 is a schematic diagram of a case in which the pressure sensor device supplied to a robot arm.

DESCRIPTION OF EMBODIMENTS

Hereinbelow, an embodiment of the invention is described with reference to the drawings.

FIG. 1 is a perspective view of a pressure sensor device according to this embodiment. Note that FIG. 1 shows a state before the device is attached to a glove.

As shown in FIG. 1, a pressure sensor device 1 is configured to include a flexible substrate 10 and a pressure sensor 20 provided on the substrate 10. The pressure sensor device 1 of this embodiment is applied to a right-hand glove and is attached to an inside of the glove to be used. The pressure sensor device 1 may be also applied to a left-hand glove.

The flexible substrate 10 has a shape that follows shapes of three fingers, the thumb, the index finger, and the middle finger. The flexible substrate 10 includes a back part 12 that is assigned to a back of the hand, fingertip parts 13 that are assigned to the fingertips, and a bridge 14 that is placed between the back part 12 and the fingertip parts 13.

The back part 12 includes a connector 31 (connection part) that is connected to a wireless transmitter module 4 (see FIG. 2). The back part 12 also includes a connection part 11a connected to an acceleration sensor 32. In addition, the fingertip part 13 for the thumb is equipped with a connection part 11b connecting the back part 12 to a microphone (sound detector) 33 for detecting sound. The bridge 14 is formed to be elongated so that it is placed along sides of the fingers.

The fingertip part 13 for the thumb is provided with a pressure sensor 20 facing a terminal segment 15 of the thumb. The fingertip part 13 for the index finger is provided with pressure sensors 20 facing a terminal segment 15 and a middle segment 16 of the index finger. The fingertip part 13 for the middle finger is provided with pressure sensors facing a terminal segment 15 and a middle segment 16 of the middle finger.

The pressure sensor 20 has a curvature in a static state. In FIG. 1, the pressure sensors 20 for the terminal segment 15 and the middle segment 16 are formed to be curved so that the sensors 20 are convex toward the back of the paper. In other words, the pressure sensors 20 in the static states are configured to be curved in directions that respectively follow bellies of the thumb, index finger, and middle finger in cross sections of their rings cut in round shape. It means that the pressure sensors 20 each has a curvature in a state in which it is left as it is (no external force is applied on it). Note that the curvature may be configured to be curved to have a single semicircle, or may be configured to be curved by combining multiple semicircles.

FIG. 2 is an external perspective view of the pressure sensor device attached to a glove.

As shown in FIG. 2, the pressure sensor device 1 is fixed to the glove 2 in a state in which the glove 2 is turned inside out. It means that the pressure sensor device 1 is fixed to the inside of the glove 2 to be used.

A pocket 2b is provided on the back of the glove 2. This pocket 2b stores a wireless transmitter module 4 (transmitter). The wireless transmitter module 4 is electrically connected via the connector 31 (see FIG. 1) to the pressure sensor 20, the acceleration sensor 32, and a microphone 33 on the flexible substrate 10. The wireless transmitter module 4 is responsible for a function of transmitting detected sensor data via a wireless connection to an edge server 103

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(see FIG. 18). Although the pressure sensor device 1 is illustrated as communicating via the wireless connection, a wired connection is also available. Details of a work management system 500 (see FIG. 18) using the pressure sensor device 1 is described below.

The pocket 2b is a so-called lidded pocket, and a lid prevents the wireless transmitter module 4 stored in the pocket 2b (pocket body) from falling out thereof even when the fingertip 2a is turned upward. The lid is not mandatory and may be dispensed with, or a fastener can be attached to close a mouth of the pocket 2b instead of the lid.

The fingertip 2a of the glove 2 is provided inside the glove with an inside pressure sensor 20, which is able to detect a fingertip pressure (load) applied on the fingertip 2a.

The microphone 33 mounted on the flexible substrate 10 is placed so as to be exposed to outside through a hole provided in a portion of the glove 2. This allows sound from outside the glove 2 to be collected. When sound passes through cloth, sound volume is attenuated or high frequency sound is filtered out, which causes degradation of the sound. In this embodiment, the hole provided for the microphone 33 is able to prevent the degradation of the sound.

FIG. 3 is a plan view showing a part of the flexible substrate used in the pressure sensor device of the embodiment. Note that FIG. 3 shows a state in which the flexible substrate 10 for the index or middle finger is flattened. FIG. 3 also shows a state without a pressure-sensitive material 21 described below.

As shown in FIG. 3, the flexible substrate 10 includes a flexible substrate base material 11 and an electrode 40 provided on the flexible substrate base material 11. The electrode 40 is provided on a front side relative to the flexible substrate base material 11 as viewed vertically with respect to the paper surface.

The flexible substrate base material 11 is formed in a film-like form using material such as polyimide or polyester and in a rectangular shape so that it is widened in a right-left direction at the terminal segment 15 and middle segment 16 of the index finger (or middle finger).

The electrode 40 is composed mainly of comb-teeth shape electrodes 41 and 42 that are pattern-printed with copper or copper foil. The comb-teeth shape electrodes 41 and 42 each is an electrode having a shape in which two electrodes are alternately patterned. In addition, the comb-teeth shape electrodes 41 and 42 are fixed to the flexible substrate base material 11 by attaching them to the material 11 via an adhesive such as epoxy resin. As shown in FIG. 8 below, each of the opposing comb teeth in the comb-teeth shape electrode 41 is configured so that it does not contact the other tooth. This is the same for the comb-teeth shape electrode 42.

The comb-teeth shape electrode 41 is located at a position corresponding to the terminal segment 15. The terminal segment 15 refers to a portion beyond the first joint of the finger. The comb-teeth shape electrode 42 is located at a position corresponding to the middle segment 16. The middle segment 16 refers to a portion between the first and second joints of the finger. Note that there is provided with a comb-teeth shape electrode at a position of the flexible substrate base material 11 corresponding to the terminal segment of the thumb.

The flexible substrate base material 11 is provided with a neck portion (necked shape) 17 formed between the positions corresponding to the terminal segment 15 and the middle segment 16, at which neck portion the flexible substrate base material 11 is formed to be narrowed. The neck portion 17 is provided with a connection electrode 43

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in a shape of a straight line that electrically connects the comb-teeth shape electrode **41** to the comb-teeth shape electrode **42**. In addition, the neck portion **17** is formed to be narrower than a portion of the flexible substrate base material **11** at a position corresponding to the terminal segment **15** and the middle segment **16**. The comb-teeth shape electrode **41** is formed to be longer than the comb-teeth shape electrode **42** in a direction of extending the finger. The comb-teeth shape electrode **41** is formed to have the same length as the comb-teeth shape electrode **42** in the width direction.

The flexible substrate base material **11** is formed asymmetrically with reference to the connection electrode (connection wiring) **43**. The comb-teeth shape electrodes **41** and **42** are also formed right-left asymmetrically with reference to the position of the connection electrode **43**. In this embodiment, the comb-teeth shape electrode on the left is configured to be larger in its electrode area than the comb-teeth shape electrode on the right.

By the way, the connection electrode **43** may be assigned along a side of a finger like the bridge **14**, but assigning the electrode **43** along a belly of the finger is found to cause less uncomfortableness when the finger is bent. The strength of the neck portion **17** is important because the first joint of the finger is bent repeatedly at a high frequency while a worker is performing a manual work. The width **D** of the neck portion **17** is determined by a trade-off between tensile strength and wearing comfortableness. The width **D** of the neck portion **17** is preferably 6.25 mm or more and 8.65 mm or less on a basis of statistics of sizes of Japanese fingers. The following is an explanation of how to obtain the width **D**.

According to publication data, "Data of Japanese Hand Dimensions (AIST: Agency of Industrial Science and Technology, URL: <https://www.airc.aist.go.jp/dhrt/hand/data/list.html>), the width of the first joint of the index finger: **d** that is measured in 327 males and 203 females is 14.9 mm in a mean value μ and 1.2 mm in a standard deviation σ . The width **D** of the neck portion **17** should be about half the width of the finger based on the trade-off between the tensile strength and the wearing comfortableness. To make width **D** match the **2a** interval of the Japanese distribution, the width **D** of the neck portion **17** should be determined by the following conditional equation:

$$(\mu - 2\sigma)/2 < D < (\mu + 2\sigma)/2.$$

Substituting the aforementioned values into the above conditional equation, the width **D** is obtained as 6.25 mm < **D** < 8.65 mm. A glove **2** of a different size (S, M, L, XL, or the like) may be prepared on a basis of this thought.

FIG. **4** is a cross-sectional view of the pressure sensor device according to the embodiment.

As shown in FIG. **4**, a pressure-sensitive material **21** is provided above the comb-teeth shape electrodes **41** and **42**. This pressure-sensitive material **21** has a characteristic that its resistance value varies depending on a load applied on it and is formed in a shape of a rectangular sheet. The pressure-sensitive material **21** is piled up so that it contacts metal surfaces of the comb-teeth shape electrodes **41** and **42**. The pressure-sensitive material **21** is, for example, an elastic material such as rubber having an insulating property mixed with carbon resin.

Further, the pressure sensor **20** is configured so that the flexible substrate base material **11**, the comb-teeth shape electrodes **41** and **42**, and the pressure-sensitive material **21**

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have a curvature in the static state of the pressure sensor **20**. The curvature radius **r** should be preferably set between 10 mm and 50 mm.

Furthermore, the pressure-sensitive material **21** is provided in a pressure-sensitive area (predetermined area) at which pressure is detected. It is desirable that a sensor configuration should be the minimum configuration required on a basis of the expected use case, because an increase in the area (number) of the pressure sensors **20** increases an amount of information detected and leads to an increase in a cost of components.

FIG. **5** is a perspective view of a state in which the pressure sensor device is attached to a finger and the finger is extended (the first and second joints are extended). FIG. **6** is a perspective view showing a state of the fingers bent from the state shown in FIG. **5**. In an actual state, the flexible substrate **10** of the pressure sensor device **1** is attached to an inside of a glove, but for explanatory convenience, the glove is omitted from FIGS. **5** and **6**. In another use, the flexible substrate **10** of the pressure sensor device **1** may be used by wrapping it directly around a finger without using a glove.

As shown in FIG. **5**, the pressure sensors **20** are formed at positions corresponding to the terminal segment **15** and middle segment **16** of the index finger **111**, and attached to a belly of the index finger **111** in a manner of winding around the belly (about halfway around the index finger **111**). The neck portion **17** is also positioned so as to overlap with the first joint **111b**. The neck portion **17** is so thin and soft that it causes little discomfort to a worker when the worker bends the index finger **111**. In addition, because the pressure sensor **20** is not located at the neck portion **17**, the pressure sensor **20** does not react even when the index finger **111** is bent, which eliminates possibility of a false reaction.

In addition, the bridge **14** is assigned along a centerline **CL** of the side of the index finger **111**. This center line **CL** corresponds to a line whose distance does not vary when the finger **111** is bent. In FIG. **4**, the distance is defined as **L** that is a distance from the first joint **111b** to a root of a base segment of the index finger **111**.

As shown in FIG. **6**, when the index finger **111** is bent, the neck portion **17** is curved at the first joint **111b**, and the bridge **14** is curved at the second joint **111c**. Even if the index finger **111** is bent at the first joint **111b** and the second joint **111c**, the center line **CL** of the side of the index finger **111a** has the distance **L**, which distance **L** does not vary from that of FIG. **4**.

In a hypothetical configuration, if the bridge **14** would be assigned along the front or back of the index finger **111**, the flexible substrate **10** would be pulled when the index finger **111** is bent, which causes not only discomfort to a wearer of this pressure sensor device, but also a pulling load on the flexible substrate **10**, which load may cause the bridge **14** to be broken or wirings placed on the flexible substrate **10** to break. However, aligning the bridge **14** with the center line **CL**, as this embodiment, may reduce the discomfort of the wearer and prevent the wiring from being broken because the flexible substrate **10** is not pulled even when the index finger **111** is bent.

FIG. **7** is an arrow view in the **S** direction of FIG. **5** and shows the index finger of the right hand viewed from the fingertip.

As shown in FIG. **7**, the flexible substrate **10** is placed in the manner of being wound around the index finger **111**. The pressure sensor **20** for the index finger **111** is configured as a pressure-sensitive area **20t** (load detection range) up to the side **111a** of the index finger **111**. This pressure-sensitive area **20t** is configured so that the side near the thumb is wider

than the side near the middle finger. In addition, the pressure sensor **20** is fixed inside the glove **2** (see FIG. **2**) in a curved state so that it winds around the finger.

This configuration allows detecting fingertip pressure generated in a manual operation such as connector insertion, as described below in FIG. **20**.

In the following, a principle of the pressure sensor **20** is described. FIG. **8** is a circuit diagram of a reading circuit reading a resistance of a pressure-sensitive material using a comb-teeth shape electrode. FIG. **9** is a cross-sectional schematic diagram of the pressure sensor. FIG. **10** is a circuit diagram of the pressure sensor. FIG. **11** is a R-F characteristics of the pressure sensor. FIG. **12** is a V-F characteristics of the pressure sensor.

As shown in FIG. **8**, the comb-teeth shape electrodes **41** and **42** are formed on the flexible substrate base material **11** (see FIG. **9**). The comb-teeth shape electrodes **41** and **42** are electrodes having shapes in which two electrodes (electrode A and electrode B) are patterned alternately. The resistance between the electrodes A and B is very high (e.g., 1 MΩ or more) because the two electrodes are electrically insulated, that is, they are patterned to be placed far enough apart so that the comb-teeth of the two electrodes do not contact each other.

As shown in FIG. **9**, the pressure-sensitive material **21** is placed above the comb-teeth shape electrodes **41** and **42**, and covered by a protective material **51**. The protective material **51** is not limited to any particular material if they have property for protecting the pressure-sensitive material **21** and the comb-teeth shape electrodes **41** and **42**, and can be, for example, a thin sheet of polyimide or the like. The pressure-sensitive material **21** has a characteristic that its resistance R is very large (e.g., more than 100 kΩ) when no load is applied and decreases (down to about 1 kΩ to 5 kΩ) when some load is applied. When a load F is applied to the pressure-sensitive material **21** placed on the comb-teeth shape electrodes **41** and **42**, the resistance R drops at a local area on which the load F is applied, which results in a drop in the resistance R between the electrodes A and B.

As shown in FIG. **10**, in an example of the reading circuit, the comb electrodes **41** and **42** and a load resistor R0 are connected in series and a voltage value V applied on the load resistor R0 is measured using a voltmeter. The voltage value V can be expressed by an equation shown in FIG. **10**.

As shown in FIG. **11**, the resistance value R varies as the load F increases. This means that the voltage value V varies with the load F as shown in FIG. **12** according to the equation shown in FIG. **10**, and therefore it can function as a pressure sensor. The comb electrodes **41** and **42** can be made to function as a pressure sensor.

Hereinbelow, description is given of an effect of a curvature bias on a sensor output using FIGS. **13A** and **13B**. FIG. **13A** is a F-R characteristics of the pressure-sensitive material without a curvature bias. FIG. **13B** is a F-R characteristics of the pressure-sensitive material with a curvature bias.

The F-R characteristic, which is a relationship between the load F and the pressure-sensitive material resistance (resistance value of the pressure-sensitive material **21**) R, depends on a composition and a production method of the pressure-sensitive material **21**. For example, FIG. **13A** shows a pressure-sensitive material whose pressure-sensitive material resistance R rapidly varies in a small range of the load F and gradually saturates. Because the reading circuit reads the resistance R, the range of variation in the

resistance R is a dynamic range of the pressure sensor, and the larger range of variation in the resistance R is more desirable.

A range of the load applied to fingertips of a worker when performing a manual work is about 100 g to 5 kg. Therefore, selecting a pressure-sensitive material with a large variation in the resistance in the above range allows improvement of detection sensitivity of the sensor. When the pressure sensor is used in a curved state to which the sensor is transitioned from a flat state, the curvature generates an internal stress in the pressure sensor. At this time, the F-R characteristic is subjected to a curvature bias load as shown in FIG. **13B**. When the curvature bias load is already applied and an additional load of 100 g to 5 kg due to a manual work is further applied on the sensor, the resistance R varies only in a significantly narrow range, which causes degradation of the detection sensitivity of the sensor. Therefore, it is important to remove the curvature bias load in order to improve the detection sensitivity of the sensor.

The next description is given of a principle of generating the curvature bias load on the pressure sensor and a structure to remove the curvature bias load. FIG. **14A** is a cross-sectional view of the pressure sensor fabricated by a flat-surface assembly. FIG. **14B** is a schematic diagram showing an internal stress when the pressure sensor of FIG. **14A** is curved. FIG. **15A** is a cross-sectional view of the pressure sensor fabricated by a curved-surface assembly. FIG. **15B** is a schematic diagram showing an internal stress when a curvature of the pressure sensor of FIG. **15A** is reduced. Note that FIGS. **14A**, **14B**, **15A**, and **15B** omit illustrations of the comb-teeth shape electrodes.

As shown in FIG. **14A**, the pressure sensor **1000** has a laminate structure in which a flexible substrate base material **1011**, a pressure-sensitive material **1021**, and a protective material **1051** are laminated. The flexible substrate base material **1011** is an inside surface and the protective material **1051** is an outside surface. Hereinbelow, the structure to remove the curvature bias load is explained.

In the laminate structure (flat-surface assembly) shown in FIG. **14A**, the pressure-sensitive material **1021** and the protective material **1051** are mounted on the flexible substrate base material **1011** that is in a flat state. This pressure sensor **1000** is an elastic material that does not have curvature in its static state, but when curved, generates a force to return into the flat state.

In the laminate structure (curved-surface assembly) shown in FIG. **15A**, the pressure-sensitive material **21** and the protective material **51** are mounted on the flexible substrate base material **11** that is in a curved state. The pressure sensor **20** of this case generates in a static state a force returning the shape of the sensor **20** to a flat surface and a force maintaining the curved shape. The magnitude of the force returning the shape to the flat surface depends on a combination of rigidities of the flexible substrate base material **11**, pressure-sensitive material **21**, and protective material **51**. However, a combination of the two forces stabilizes the shape in a slightly flatter state (this state is called "stable curvature"). This results in the pressure sensor **20** having a curvature corresponding to the stable curvature in the static state. This pressure sensor **20** is an elastic body that exerts a force to return the state of the sensor **20** to the stable curvature when the sensor **20** is curved.

FIG. **14B** shows with solid arrows the internal stress applied on the pressure sensor **1000** when it is curved so that it is wound around a fingertip. FIG. **15B** shows with solid arrows the internal stress applied on the pressure sensor **20** when it is curved so that it is wound around the fingertip.

As shown in FIG. 14B, when the pressure sensor **1000** is curved, a compressive stress is generated in the flexible substrate base material **1011** that is the inside surface, and a tensile stress is generated in the protective material **1051** that is the outside surface. This causes the pressure-sensitive material **1021** that is placed intermediate to generate a compressive stress generated in the directions perpendicular to the inside and outside surfaces. This stress in the perpendicular directions is called an interlaminar stress. The interlaminar stress due to the compression is the cause of the curvature bias load.

As shown in FIG. 15B, when the curvature is eliminated for the pressure sensor **20**, which has the curvature (the curvature along the circular cross section of the finger (finger belly)) in the static state, the tensile stress is generated in the flexible substrate base material **11** that is the inside surface, and the compressive stress in the protective material **51** that is the outside surface. This results in the tensile interlaminar stress generated in the intermediate pressure-sensitive material **21**. This tensile interlaminar stress corresponds to a negative load being applied to the pressure-sensitive material **21**. This means that the pressure sensor **20**, which has the curvature in its static state as shown in FIG. 15A, has the curvature bias load eliminated by eliminating the curvature.

On a base of the above-described thought, assuming a curvature radius r for using the sensor by wrapping it around a fingertip, and mounting the pressure-sensitive material **21** with a curvature radius even more curved than the curvature radius r , allows not only preventing a curvature bias load from being generated, but also eliminating the curvature bias load.

FIG. 16 is curvature characteristics of the pressure sensors fabricated by a flat-surface assembly and a curved-surface assembly. Note that FIG. 16 shows a case of the flat-surface assembly using the graph connecting the white circles, and a case of curved-surface assembly using the graph connecting the black circles. The pressure sensor **20** made by the curved-surface assembly was fabricated using a cylindrical jig **60** (jig with a curved surface, see FIG. 17) having a curvature radius of 20 mm. The resistance R of the pressure-sensitive material **21** was measured by wrapping and fixing each pressure sensor **20** around an object with a predetermined curvature radius. At this time, no load was applied from the outside of the surface of the pressure sensor **20**. The manufacturing method of the pressure sensor **20** is described below.

The pressure sensor **20** shows a behavior in which its resistance value R decreases when a load is applied, and therefore the larger resistance value R is desirable because, the larger resistance value R causes the range of variation in the resistance value R to be wider and the detection sensitivity to be higher. As shown in FIG. 16, when the curvature radius of the pressure sensor is small, that is, the pressure sensor is wrapped around an object having a smaller curvature radius or a thinner object, the curvature bias load causes both of the pressure sensors made by either the flat-surface assembly and the curved-surface assembly to be small in the resistance values R and to be narrow in the sensor operating ranges. In another viewpoint, the pressure sensor **20** of the curved-surface assembly shows the large resistance value R of 500 k Ω or more in the area where the curvature radius is 20 mm or more, which indicates that a wide sensor operating range can be kept. In addition, the pressure sensor of the curved-surface assembly shows the larger resistance value R than that of the flat-surface assembly over all curvature radii. This result shows that a negative

load is applied so that the curvature bias is removed when the curved state in the static state of the curved pressure sensor **20** is eliminated (i.e., widened) as shown in FIG. 15B. Based on the above study, it is considered that the curvature bias can be effectively removed by adding the curvature in a static state by using a cylindrical jig **60** having a curvature radius smaller than a curvature radius when the pressure sensor **20** is used (curvature radius when wound around the finger).

FIG. 17 is a process diagram showing a manufacturing method of the pressure sensor device.

First, as shown in the first figure from the top of FIG. 17, the flexible substrate **10** (the flexible substrate base material **11** plus the comb-teeth shape electrodes **41** and **42**) is fixed to the cylindrical jig **60** with a double-sided tape and the like. The comb-teeth shape electrodes **41** and **42** of the flexible substrate **10** are facing outward. The above step keeps the flexible substrate **10** in a curved state (state with curvature). Then, as shown in the second figure from the top in FIG. 17, the pressure-sensitive material **21** (pressure-sensitive sheet) is placed and stuck on the comb-teeth shape electrodes **41** and **42** of the flexible substrate **10** kept in the curved state. This ensures the pressure-sensitive material **21** also to be kept in a curved state (a state with the curvature). Then, as shown in the third figure from the top in FIG. 17, the protective material **51** (protective sheet) is stuck on the pressure-sensitive material **21** that is still in the curved state. The protective material **51** consists mainly of, for example, a polyimide tape. Then, as shown in the fourth figure from the top of FIG. 17, the sheet-like material that includes the pressure-sensitive material **21** and the protective material **51** laminated on the flexible substrate **10** is removed from the cylindrical jig **60**. This results in the pressure sensor **20** (pressure sensor device **1**) having the curvature in the static state.

The above-mentioned manufacturing method of the pressure sensor **20** (pressure sensor device) is only one example and is not limited to this embodiment. For example, the following is another manufacturing method of the pressure sensor **20**: pouring ink containing carbon particles (flowable elastic resin) onto the comb-teeth shape electrodes **41** and **42** on the curved flexible substrate **10**; and molding the pressure-sensitive material **21** by heating. This method allows a productivity of the pressure sensor **20** to be improved comparing with the manual manufacturing method described above.

FIG. 18 is a configuration diagram of a work management system managing a work performed by a worker

As shown in FIG. 18, the work management system **500** includes a sensor built-in glove **101** (a pressure sensor device), and an edge server **103** (status manager and controller) that performs status management on a basis of a voltage value V transmitted from a transmitter **102** provided in the sensor built-in glove **101**. The transmitter **102** transmits the voltage value V measured by the sensor built-in glove **101**.

The sensor built-in glove **101** includes: the flexible substrate base material **11** (see FIG. 9); the comb-teeth shape electrodes **41** and **42** (see FIG. 9) with exposed metal surfaces formed on a predetermined area on the flexible substrate base material **11**; the pressure-sensitive material **21** (see FIG. 9) that is provided above the comb-teeth shape electrodes **41** and **42**, varies in the resistance value R depending on the load, and has the curvature in the static state; and the transmitter **102**.

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A worker **100** wears the sensor built-in glove **101** to perform a manual work. The manual work includes assembly, equipment inspection, work using a tool, checking, and the like.

The sensor built-in glove includes the pressure sensor **20** that detects a pressure applied to the fingertip, an acceleration sensor **32** and a gyroscopic sensor **34** that detect movement of the hand, and a microphone **33** that detects work sounds produced by the fingertip.

Sensor data detected by the sensors incorporated in the glove **101** (pressure sensor **20**, acceleration sensor **32**, microphone **33**, gyroscopic sensor **34**) is transmitted through a wireless channel of the transmitter **102** to the edge server **103** that is the status manager. The channel for transmitting the sensor data is not limited to be wireless, but may also be wired. The edge server **103** is an electronic device equipped with a CPU, such as a PC or a smartphone.

The edge server **103** performs arithmetic operation on the sensor data and provides feedback to the worker, as necessary. The feedback includes, for example, alerting the worker with a patrol lamp, and the like, when the edge server **103** determines that the worker is performing dangerous work. In one of other cases, for example, if a behavior of the assembly worker deviates from a normal behavior, the server **103** alerts the worker to redo the assembly in order to improve a quality of an assembled product.

The edge server **103** transmits data through a network **104** to a central server **105**. The edge server **103** performs functions of storing sensor data and collaborating with other systems. Note that the arithmetic operation on the sensor data may be performed by either the edge server **103** or the central server **105**. In addition, the edge server **103** and the central server **105** may be the same device. The edge server **103** and the central server **105** may also communicate directly with each other via wired or wireless means without intermediation through the network **104**.

FIG. **19** is a flowchart showing an operation of the work management system.

As shown in FIG. **19**, first in step **S200**, sensor data from the sensors incorporated in the sensor built-in glove **101** (see FIG. **18**) is measured. The measurement of the sensor data is repeated at a specified sampling interval (e.g., every one second, every 0.01 second, or the like). Alternatively, the measurement may be triggered by an external input to the system that the worker has performed a series of tasks. For example, the system detects movements of the worker **100** using a camera and measures the sensor data on a timing when detecting an end of the work from video image.

In step **S201**, the edge server **103** (or the central server **105**) performs arithmetic operation on the sensor data. The arithmetic operation includes resampling, digital filtering, noise removal, calculation of statistical values (such as mean, variance, standard deviation, and the like), spectrum calculation, pattern matching, regression analysis, and classification generalized as machine learning (such as support vector machine, logistic regression, random forest, k-nearest neighbor algorithm, etc.).

In step **S202**, the edge server **103** determines whether or not feedback to the worker is necessary in view of a result of the calculation. If the edge server **103** determines that the feedback is necessary for the worker **100** (**S202**, YES), it proceeds to step **S203** to issue the feedback to the worker **100**. If the edge server **103** determines that the feedback is unnecessary for the worker **100** (**S202**, NO), it proceeds to step **S204** to record the sensor data. The edge server **103** also proceeds to step **S204** to record the sensor data even after issuing the feedback to the worker (**S203**). In this way, the

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edge server **103** (or the central server **105**) records the sensor data regardless of whether or not the feedback is issued (**S204**).

In order to avoid a huge amount of recorded sensor data, it is assumed that only some sensor data is recorded. For example, when it is determined that feedback is necessary, the sensor data is of high importance because the worker may be performing non-regular behavior such as a dangerous manual work. Recording only the sensor data when it is determined that the feedback is necessary is an effective way to reduce the amount of the recorded sensor data.

Hereinbelow, an example of utilizing the work management system **500** is described with reference to the drawings.

FIG. **20** is a perspective view of an example of a connector insertion work. For convenience of explanation, FIG. **20** shows a state of the glove being removed.

As shown in FIG. **20**, the connector insertion work is a work of inserting the connector **200** into the insertion target **201**. If the connector is not properly inserted, the connector **200** may have faulty electrical contact, or the connector may come off due to vibration. Further, in a case in which the insertion target **201** is an automobile, if the connector **200** is disconnected after the product is shipped, it may lead to an accident. As described above, it is important to the work of inserting the connector to ensure a quality of the connector-inserted state.

However, the connector insertion work is difficult to automate with robots, and is often performed manually by an assembly worker. This system (work management system) can be effectively utilized to ensure the quality of the connector insertion work that is manually performed.

The pressure sensor device **1** applied to this system is provided with pressure sensors **20** and the microphone **33**, which sensors **20** are assigned at tips of the thumb **110** and the index finger **111** of the worker's hand **202** for grasping the connector **200**. The pressure sensors **20** detect that the worker has grabbed the connector **200**. The microphone **33** detects the interlocking sound emitted when the connector **200** is inserted into the insertion target **201**. In addition, the pressure sensor **20** is positioned at a location opposite the terminal segment **15** of the thumb **110** and locations opposite the terminal segment **15** and middle segment **16** of the index finger **111**.

Analyzing the sensor data obtained from the pressure sensor **20** and the microphone **33** allows determination on whether or not the connector insertion work has been properly performed. If the connector insertion work is determined to be inappropriate (YES in **S202** of FIG. **19**), the system can provide feedback to the worker by means of a patrol lamp (**S203** in FIG. **19**) to instruct him/her to redo the insertion work. In this way, the quality of the connector insertion work can be ensured.

In the work example shown in FIG. **20**, description is given as an example using the pressure sensor device **1** provided with a single microphone **33**. However, in addition to the microphone **33** near a drive unit, the pressure sensor device **1** may be configured to have another microphone added at a position far from the drive unit.

FIG. **21A** is a perspective view showing a shape of a hand holding a connector and shows an example of a shape of the hand holding a LAN cable connector **200**. A way to hold the connector **200** varies from person to person. The way of holding the connector **200** shown in FIG. **21A** is using the belly of the thumb **110** and the side of the index finger **111** to grasp the connector. In this way of holding, in order to detect a force of the index finger **111** when it inserts the connector **200**, the pressure-sensitive area (load detection

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range of the pressure sensor **20** of the index finger **111** needs to cover the side **111a** of the index finger **111** (see FIG. 21A) as the pressure-sensitive area **20t** in addition to the pressure-sensitive area **20s** on the belly of the index finger **111**, as shown in FIG. 21B. In addition, including both of the terminal segment **15** and the middle segment **16** of the index finger **111** into the pressure-sensitive area allows covering various grasping ways that differ depending on personal characters.

FIG. 22 is a perspective view of an example of a work using a power tool. Note that FIG. 22 shows an example of a work using an electric screwdriver as a usage case applying the pressure sensitive device

As shown in FIG. 22, when grasping the electric screwdriver **300**, the grip **301** is held by the thumb, middle finger, ring finger, and little finger, and the index finger is placed on the trigger **302**. Grasping the electric screwdriver **300** in this way applies a load to the pressure sensors **20** of the terminal segment **15** of the thumb (see FIG. 1) and the pressure sensors **20** of the terminal segment **15** (see FIG. 1) and the middle segment **16** (see FIG. 1) of the middle finger. Because there are individual differences in a way of pulling the trigger **302**, it is desirable to use both the terminal segment **15** and the middle segment **16** of the index finger as the pressure-sensitive areas.

When working with such an electric screwdriver **300**, pulling the trigger **302** applies pressure to the index finger, which causes the pressure sensor **20** on the index finger to capture the action of pulling the trigger **302**. In addition, when the trigger **302** is pulled and the driver at a tip end is rotated and driven, a driving sound is emitted and captured by the microphone **33**. In this way, managing the sensor data of the pressure sensor **20** and the microphone **33** can confirm the quality of the work using the electric driver **300**.

FIG. 23 is a perspective view showing an example of a work for opening/closing a valve. The gloves are omitted in FIG. 23.

As shown in FIG. 23, a valve work is a process of opening and closing a valve **401** equipped in a cylinder **400**. Regulating a closing level of the valve **401** adjusts a flow rate of gas and opens/closes a gas channel. The valve work is typically performed, for example, in an inspection of a plant, a semiconductor manufacturing equipment, and the like. If the valve work is not performed properly, a reverse flow of gas or mixed flow of gas may occur. The valve **401** can be replaced by a mechanical device such as a solenoid valve, but valves to be manually opened and closed are also widely used.

This system (work management system) can be utilized to keep a work record of the worker's proper performance in the valve work. This system includes the pressure sensors **20** of the thumb **110**, the index finger **111**, and the middle finger **112**, which sensors **20** detect the force of the fingers grasping the valve **401**, and a gyroscopic sensor **34** that detects a rotational motion of the hand **3**,

A sequence of operations for closing the valve **401** is as follows. First, the valve **401** is grasped and rotated by hand. Once the valve **401** is closed to some extent, it is stopped rotating. More force is applied in a direction of rotation (in a direction of the valve **401** closing). Further force is applied in the direction in which the valve **401** is closed to confirm that the valve **401** is completely closed (confirmation work).

Using this system allows the gyroscopic sensor **34** to detect the rotation of the hand **3**, and the pressure sensor **20** to detect the action of applying force with the fingers in the confirmation work. In this way, the record of the work performed to close the valve **401** can be stored. The strength

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of the force applied in the confirmation work with the fingers is difficult to quantify using the conventional art. However, using this system allows the strength of the force to be quantified as the sensor data of the pressure sensors **20**. This way allows tacit knowledge in a work field to be visualized.

As explained above, the pressure sensor device **1** of this embodiment includes the flexible substrate base material **11** having flexibility, the comb-teeth shape electrodes **41** and **42** that have exposed metal surfaces formed in a predetermined area on the flexible substrate base material **11**, and the pressure-sensitive material **21** that is provided on the comb-teeth shape electrodes **41** and **42** and has the resistance value **R** varying with the load and the curvature in the static state. This makes it possible to reduce or eliminate the curvature bias to improve the detection sensitivity and to mitigate a degree of a false detection.

In this embodiment, the pressure sensor device **1** is fixed to the inside of the glove. This enables stably holding the pressure sensor device **1** and easily attaching and removing the pressure sensor device **1**. In addition, because the pressure sensor device **1** does not directly contact an object, the pressure sensor device **1** can be suppressed from being deteriorated.

In this embodiment, the flexible substrate base material **11** is equipped with the connection part **11b** connecting with the microphone **33**, the connection part **11a** connecting with the acceleration sensor **32**, and the connector **31** connecting with the wireless transmitter module **4**. According to this configuration, equipping the microphone **33** allows minute work sounds near a hand to be captured. Further, equipping the acceleration sensor **32** allows capturing the movement of a hand and other objects. Thus, integrally using multiple types of sensors enables capturing more detailed information on the manual work. Furthermore, equipping the wireless transmitter module **4** allows easily transmitting the detected sensor data (information) such as the detected pressure to the computing processor of the edge server **103** and the like. In addition, the wireless transmitter module **4** is able to improve usability of the system compared to the wired system.

In this embodiment, the sides of the terminal segment **15** and the middle segment **16** of the index finger **111** are pressure-sensitive areas **20t** as predetermined areas (see FIGS. 7, 20, 21A, and 21B). This allows the connector insertion work to be properly managed.

In this embodiment, the flexible substrate base material **11** is fixed to the glove **2** so that the bridge **14** of the base material **11** extending from the finger to the back of the hand is placed on the side of the finger (see FIG. 7). This helps to alleviate discomfort when the wearer of the glove **2** bends and stretches the finger.

In this embodiment, the pressure-sensitive area **20s** includes the terminal segment **15** and middle segment **16**, and the flexible substrate base material **11** includes the neck portion **17** formed with a narrow width of the flexible substrate base material **11** at a position overlapping with the first joint **111b** that connects the terminal segment **15** with the middle segment **16** (see FIG. 3). This alleviates a degree of a false reaction of the sensor caused by bending the fingers and also reduces the discomfort of the worker due to bending and stretching the fingers.

In this embodiment, the flexible substrate base material **11** is provided with the comb-teeth shape electrodes **41** and **42** formed asymmetrically with reference to the connection electrodes **43** that electrically connect the terminal segment **15** to the middle segment **16** (see FIG. 3). This configuration allows the side **111a** of the index finger **111** to be used as the

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pressure-sensitive area **20t**, making it possible to perform the connector insertion work, and the like in a suitable manner.

In this embodiment, a width D of the neck portion **17** is 6.25 mm or more and 8.65 mm or less (see FIG. 3), which allows a majority of workers to feel less discomfort when they bends and stretches the fingers.

In this embodiment, the radius of curvature r in the static state (abbreviated as “curvature radius”) of the pressure-sensitive material **21** is set to 10 mm to 50 mm (see FIG. 4). Setting the curvature radius r in such a range allows the embodiment to be utilized not only for hands but also for curved surfaces of various objects such as a foot.

In this embodiment, the pressure-sensitive material **21** is attached on the finger in a state in which the curvature radius r in its static state is further smaller than the curvature radius of the fingertip (curved surface of a target on which the pressure-sensitive material **21** is attached). This configuration not only prevents the curvature bias load from being generated, but also removes the curvature bias load to obtain a highly sensitive pressure sensor device **1**.

In addition, the work management system **500** of this embodiment includes the pressure sensor device **1** including the flexible substrate base material **11** having flexibility, the comb-teeth shape electrodes **41** and **42** that are formed in a predetermined area on the flexible substrate base material **11** and have exposed metal surfaces, and the pressure-sensitive material **21** that is provided on the comb-teeth shape electrodes **41** and **42**, and has the resistance value R varying with the load and the curvature in the static state; and the edge server **103** (or the central server **105**) that performs status management based on the voltage value V transmitted by the wireless transmitter module **4**. This configuration allows the manual work in the factory assembly processes, equipment inspection work, and the like to be digitized with sensors and the work management system **500** to be applied for recording work histories, preventing work errors, and extracting proficient skills.

In the work management system **500**, the edge server **103** (or the central server **105**) provides a feedback on a work status according to the result of the status management, and performs a warning process based on the feedback (see S202 and S203 in FIG. 19). This configuration may prevent work errors from being overlooked.

In the manufacturing method of the pressure sensor device of this embodiment, a flexible substrate base material **11** on which comb-teeth shape electrodes **41** and **42** are formed is attached to the cylindrical jig **60** having a curved surface by wrapping the material **11** around the cylindrical jig **60**. The pressure-sensitive material **21** whose resistance value varies depending on a load is formed above the comb-teeth shape electrodes **41** and **42** in the curved state. After adhering the protective material **51** over the pressure-sensitive material **21**, the flexible substrate base material **11** including the comb-teeth shape electrodes **41** and **42**, the pressure-sensitive material **21**, and the protective material **51** is removed from the cylindrical jig **60**. According to this method, a pressure sensor device with improved detection sensitivity and excellent capability of mitigating a degree of a false reaction can be manufactured in a simple process.

Note that the present invention is not limited to the aforementioned embodiments and may include various modifications. For example, in this embodiment, description is given of a case in which the pressure sensor device **1** is mounted on the glove **2**. However, the pressure sensor device may be applied to a sock. Specifically, the pressure sensor device can be applied to a testing of a brake in a

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manufacturing process of automobiles, and configured to use a base of a toe as the pressure-sensitive area.

Further, the pressure sensor device can also be applied to a robot arm **700** instead of the human hand. As shown in FIG. 24, pressure sensor devices with the pressure sensors **20** are fixed to two finger parts provided in the robot arm **700**. Applying the pressure sensor device to a robot in this way enables various types of work management to be done for various lots of works. In addition, this embodiment can be configured at a lower cost than equipping the pressure sensor on the robot itself.

Another embodiment includes a flexible substrate base material **11** shaped to follow a hand, comb-teeth shape electrodes **41** and **42** with exposed metal surface formed in the pressure-sensitive area on the flexible substrate base material **11**, and the pressure-sensitive material **21** that is provided on the comb electrodes **41** and **42** and has the resistance value R varying with a load. The pressure-sensitive area includes areas overlapping the terminal segment **15** and middle segment **16** of a finger. In the flexible substrate base material **11**, the first joint **111b** connecting the terminal segment **15** with the middle segment **16** is provided with the neck portion (necked shape) **17** at which the width D of the flexible substrate base material **11** is narrow. This configuration is able to eliminate discomfort caused to a worker when the finger is bent.

Also, in this embodiment, description is given as an example of a case in which the pressure sensor **20** is mounted on the thumb **110**, index finger **111**, and middle finger **112**. However, the pressure sensor **20** may be mounted on two fingers of the thumb **110** and index finger **111** or on four or more fingers.

This embodiment is described using as an example the pressure sensor device **1** with the curvature to follow along the cross-section of a finger being cut into a circle. However, this embodiment is not limited to such a configuration. For example, because a finger is curved so as to taper from its first joint **111b** toward its fingertip, the pressure sensor device may be configured to have a curvature in a static state following along the tapering. This configuration is able to reduce or eliminate the curvature bias to improve the detection sensitivity and to mitigate the degree of the false detection.

In addition, this embodiment is described as an example using the case in which the pressure sensor device **1** is fixed to a glove. However, the pressure sensor device **1** may be fixed directly to the worker's hand.

REFERENCE SIGNS LIST

- 1: Pressure sensor device
- 2: Glove
- 2a: Fingertip
- 2b: Pocket
- 3: Worker's hand
- 4: Wireless transmitter module (transmitter)
- 10: Flexible substrate
- 11: Flexible substrate base material
- 11a, 11b: Connection part
- 12: back part
- 13: Fingertip
- 14: Bridge
- 15: Terminal segment
- 16: Middle segment
- 17: Neck portion (necked shape)
- 20: Pressure sensor
- 20s, 20t: Pressure-sensitive area (predetermined area)

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21: Pressure-sensitive material
 31: Connector (connection part)
 32: Acceleration sensor
 33: Microphone
 34: Gyro sensor
 40: Electrode
 41, 42: Comb-teeth shape electrode
 43: Connection electrode (connection wiring)
 51: Protective material
 60: Cylindrical jig (jig with curved surface)
 100: Worker
 110: Thumb
 111: Index finger
 111a: Side of index finger
 111b: First joint of index finger
 111c: Second joint of index finger
 112: Middle finger
 101: Gloves with built-in sensor (pressure sensor)
 102: Transmitter
 103: Edge server (status manager)
 104: Network
 105: Central server
 200: Connector
 201: Insertion target
 202: Worker's hands
 203: Pressure-sensitive area
 300: Electric driver
 301: Grips
 302: Triggers
 400: Cylinder
 401: Valve
 500: Work management system
 CL: Center line
 r: Curvature radius
 R: Resistance
 The invention claimed is:
 1. A pressure sensor device mounted on a target for detecting pressure, the pressure sensor device comprising:
 a flexible substrate base material having flexibility;
 an electrode having an exposed metal surface and formed in a predetermined area on the flexible substrate base material; and
 a pressure sensor including a pressure-sensitive material that is provided on the electrode and configured to vary in a resistance value thereof depending on an amount of a load,
 wherein the pressure sensor including the pressure-sensitive material is configured to be curved and then laminated so as to have a curvature in a static state, and wherein a radius of a curvature of the pressure sensor device in the static state is smaller than a radius of a curvature of a curved surface of the target, the curvature of the pressure sensor device increasing the resistance value of the pressure-sensitive material for at least one of: reducing a curvature bias load on the pressure sensor, increasing a sensitivity of the pressure sensor, or reducing a degree of false detection.
 2. The pressure sensor device according to claim 1, wherein
 the target is a glove and the pressure sensor device is configured to be mounted on an inside of the glove.
 3. The pressure sensor device according to claim 2, wherein
 the flexible substrate base material is configured to be fixed to the glove so that a bridge extending from a finger to a back of a hand, when inside the glove, is disposed on a side of the finger.

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4. The pressure sensor device according to claim 1, wherein
 the flexible substrate base material is equipped with a connection part configured to connect the pressure sensor device with at least one of a microphone, an acceleration sensor, and a transmitter.
 5. The pressure sensor device according to claim 1, wherein
 the predetermined area on the flexible substrate base material includes a first pressure sensitive area for corresponding to a terminal segment of a finger and a second pressure sensitive area for corresponding to a middle segment of the finger.
 6. The pressure sensor device according to claim 5, wherein
 the first pressure-sensitive area is configured for overlapping the terminal segment of the finger and the second pressure-sensitive area is configured for overlapping the middle segment of the finger, and
 the flexible substrate base material is configured to include a neck portion between the first and second pressure-sensitive areas, and at which a width of the flexible substrate base material is formed to be narrowed at a position for overlapping a joint of the finger connecting the terminal segment with the middle segment of the finger.
 7. The pressure sensor device according to claim 6, wherein
 the flexible substrate base material is formed so that the electrode is right-left asymmetrical with reference to a connection wiring that is configured to electrically connect between portions of the flexible substrate base material corresponding to the terminal segment and the middle segment.
 8. The pressure sensor device according to claim 6, wherein
 the width of the neck portion is 6.25 mm or more and 8.65 mm or less.
 9. The pressure sensor device according to claim 8, wherein
 the radius of the curvature of the pressure sensor device in the static state is 10 mm to 50 mm.
 10. A work management system comprising:
 a pressure sensor device mounted on a target, the pressure sensor device including:
 a flexible substrate base material having flexibility;
 an electrode configured to have an exposed metal surface formed in a predetermined area on the flexible substrate base material;
 a pressure sensor including a pressure-sensitive material that is provided on the electrode and configured to vary in a resistance value thereof depending on an amount of a load; and
 a transmitter that transmits at least a voltage value measured by the pressure-sensitive material; and
 a status manager configured to perform status management on a basis of the voltage value transmitted from the transmitter,
 wherein the pressure sensor including the pressure-sensitive material is configured to be curved and then laminated so as to have a curvature in a static state, and wherein a radius of a curvature of the pressure sensor device in the static state is smaller than a radius of a curvature of a curved surface of the target, the curvature of the pressure sensor device increasing the resistance value of the pressure-sensitive material for at least one of: reducing a curvature bias load on the

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pressure sensor, increasing a sensitivity of the pressure sensor, or reducing a degree of false detection.

11. The work management system according to claim **10**, wherein

the status manager is configured to perform a feedback 5
about a work state according to a result of the status
management; and
to perform a warning based on the feedback.

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